

Bacteriophage infection drives loss of β -lactam resistance in methicillin-resistant *Staphylococcus aureus*

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eLife Assessment

The manuscript explores how bacterial evolution in the presence of lytic phages modulates β -lactams resistance and virulence properties in methicillin-resistant *Staphylococcus aureus* (MRSA). The work is **useful** as it identifies underlying mutations that may confer sensitivity to β -lactams and alter virulence properties. While the findings are generally **convincing**, additional experiments linking how particular mutations regulate phenotypic changes are required to improve the work mechanistically.

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Abstract

Bacteriophage (phage) therapy has been proposed as a means to combat drug-resistant bacterial pathogens. Infection by phage can select for mutations in bacterial populations that confer resistance against phage infection. However, resistance against phage can yield evolutionary trade-offs of biomedical use. Here we report the discovery of staphylococcal phages that cause different strains of methicillin-resistant *Staphylococcus aureus* (MRSA) to become sensitized to β -lactams, a class of antibiotics against which MRSA is typically highly resistant. MRSA cells that survive infection by these phages display significant reductions in minimal inhibitory concentration against different β -lactams compared to uninfected bacteria. Phage-treated MRSA further exhibited attenuated virulence phenotypes in the form of reduced hemolysis and clumping. Sequencing analysis revealed that the different MRSA strains evolved unique genetic profiles during infection. These results suggest complex evolutionary trajectories in MRSA during phage predation and open up new possibilities to reduce drug resistance and virulence in MRSA infections.

Staphylococcus aureus is a Gram-positive bacterium found on the skin and mucosa of humans and animals. It causes a wide range of human diseases, from endocarditis to pneumonia.¹ Methicillin-resistant *S. aureus* (MRSA) is a form of *S. aureus* that displays high levels of drug resistance towards β -lactam antibiotics, such as penicillins and cephalosporins. β -lactams act by inhibiting the activity of transpeptidase enzymes during peptidoglycan synthesis in bacterial cell walls.² β -lactams are one of the most commonly prescribed drug classes, and many are designated as “Critically Important” antimicrobials by the World Health Organization.³

A chief mediator of β -lactams resistance in MRSA is the SCCmec cassette, a mobile genetic element that carries the resistance gene *mecA*. *MecA* encodes for the penicillin-binding protein 2A (PBP2a), a transpeptidase that has a low affinity for β -lactams.⁴ This lower affinity permits PBP2a to participate in peptidoglycan synthesis even in the presence of β -lactams, ultimately resulting in cell survival. In addition to *mecA*, numerous MRSA strains carry β -lactamases, such as BlaZ, that degrade β -lactams, thus further contributing to drug resistance. MRSA infections pose a considerable public health risk as they are notoriously difficult to treat and are widespread in communities and hospital settings. In 2019, MRSA alone accounted for more than 100,000 deaths attributable to drug-resistant infections worldwide.⁵ Developing novel solutions to combat MRSA is a major focus in academia and industry.

Bacteriophages (phages) are viruses that infect and kill bacteria, posing one of the greatest existential threats to bacterial communities. Some estimates suggest that 40% of all bacterial mortality worldwide is caused by phage predation.⁶ Due to this lethality, phage therapy has been proposed as an attractive alternative for combating drug-resistant bacterial infections. In phage therapy, lytic phages are administered to kill the bacterial pathogen(s) causing an infection. Phages offer certain advantages over traditional antibiotics: they are highly specific for their hosts by reducing off-target killing; they self-amplify and evolve, enabling the rapid generation of new phage variants with improved activities; and they are generally regarded as safe, based on extremely rare cases of toxicity in animals and patients.⁷ Indeed, against staphylococcal infections, over a dozen promising case studies and clinical trials have been reported in the past decade.⁸ Despite these advances, routine use of phage therapy is still met with challenges.

Chief among these challenges is the inevitable rise of phage resistance as phage predation exerts a strong selective pressure on bacterial populations. According to one analysis that focused on phage therapy outcomes, resistance against phage evolved in 75 percent of human clinical cases in which the evolution of resistance was monitored.⁹ Mutations represent a chief pathway by which bacteria evolve resistance against phage. To date, the best characterized phage resistance mutations strains involve alterations of the cell surface that affect phage attachment. For most lytic phages, the first step of their infection involves attachment of the phage particle to a receptor molecule (or multiple molecules) on the cell surface; these receptors are often proteins or sugar moieties, which are recognized by phage proteins.^{8,10,11} Mutations in these receptor molecules can prevent phage binding, leading to resistance. For example, in *Escherichia coli*, mutations in the cell-wall protein LamB confer resistance against lambda phage infection, while in *S. aureus*, mutations that modify the glycopolymers in the cell wall, such as wall teichoic acid (WTA), have been shown to limit the infection by phages.^{11,12} Complicating the picture, studies have revealed that a plethora of additional host mechanisms, including dedicated anti-phage defense systems, can impact the evolution of resistance against phage.^{13,14} These problems highlight the importance of developing phage treatment strategies that minimize or capitalize on the evolution of phage resistance.¹⁵

A unique aspect of phage therapy is the possibility to exploit evolutionary trade-offs to combat resistant pathogens. A genetic trade-off is defined as an evolved trait that confers a fitness advantage against a particular selective pressure at the expense of reduced fitness against an unselected pressure. Across many different species of bacteria, such fitness trade-offs are known to occur between phage resistance and antibiotic resistance (**Figure 1A**). For example, selection by a lytic phage PYO^{Sa} on the methicillin-sensitive *S. aureus* Newman selected for mutations in the *femA* aminoacyltransferase, leading to increased resistance against phage infection but reduced resistance towards antibiotics.¹⁶ These studies raise the question of whether such phage/antibiotic resistance trade-offs can be induced in actual MRSA strains, which would have significant therapeutic implications. Employing phages to drive the evolution of such beneficial trade-offs could serve as a means to reduce drug resistance. However, very little is known about how frequently these trade-offs occur in different bacterial species, or through what molecular mechanisms they work.

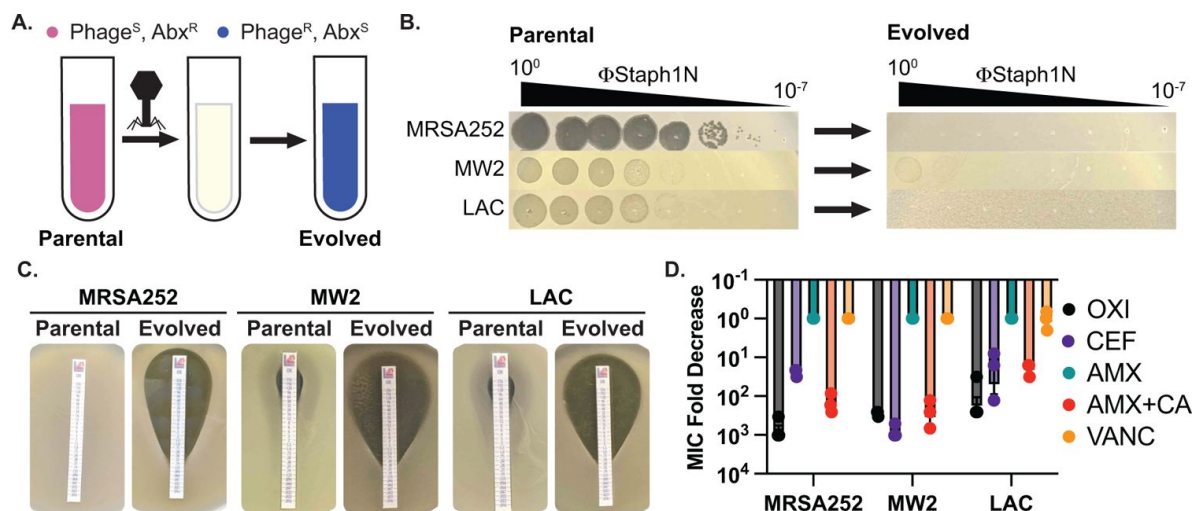


Figure 1.

Infection by bacteriophage $\Phi\text{Staph1N}$ drives the loss of β -lactam resistance in MRSA.

A. Schematic of the experimental setup. Drug-resistant (Abx^R), phage-sensitive (Phage^S) bacterial cultures are infected with phage. The population of infected cells is passaged and allowed to recover. The surviving cell population is resistant to phage infection (Phage^R) but has evolved sensitivity to antibiotics (Abx^S). **B.** $\Phi\text{Staph1N}$ infects MRSA strains MRSA252, MW2, and LAC (left panel). Following infection with $\Phi\text{Staph1N}$, evolved cultures of the three MRSA strains are resistant to $\Phi\text{Staph1N}$ (right panel). **C.** $\Phi\text{Staph1N}$ -treated, evolved MRSA strains show significant loss of resistance against oxacillin, compared to the parental strains. Loss of resistance is indicated by the area of bacterial clearance surrounding the antibiotic resistance strip. **D.** $\Phi\text{Staph1N}$ treatment causes loss resistance against different β -lactams. Plotted are the fold reductions of minimal inhibitory concentration (MIC) between treated and mock-treated cells. OXI = oxacillin; CEF = cefazolin; AMX = amoxicillin; AMX+CA = amoxicillin & clavulanic acid; VANC = vancomycin.

Here we investigate how infection by staphylococcal phages causes three different MRSA strains to evolve sensitivity to different types of β -lactam antibiotics. Interestingly, despite being treated by the same phage, the loss of resistance phenotype is mediated by mutations distinct to each MRSA strain. In addition to resistance, phage infection also causes a reduction in hemolysis and clumping, which are key virulence phenotypes in MRSA. Finally, we isolated a mutant version of the phage that displays enhanced killing, improved resistance reduction, and reduced virulence phenotypes across all three MRSA strains.

Results

Identification of Φ Staph1N with activity against multiple MRSA strains

We subjected three MRSA strains – MRSA252 (USA200), MW2 (USA400), and LAC (USA300) – to a panel of staphylococcal phages using plaquing assays ([Table S1](#), [Figure S1](#)). All three MRSA strains are pathogenic isolates that have been implicated in human infection and disease and have been used as representative examples for studying MRSA.^{17–19} Of the phages tested, one phage, Φ Staph1N, was able to form distinct plaques on all three MRSA strains ([Figure 1B](#)). Φ Staph1N belongs to the myovirus family of lytic staphylococcal phages, with a genome size of ~138 kB.^{20,21}

Despite its ability to infect all three MRSA strains, Φ Staph1N infection was often unable to effectively lyse MRSA cultures. For example, both MW2 and LAC displayed incomplete lysis in liquid culture at multiplicities of infection (MOIs) of 0.1 or lower ([Figure S2](#)). At an MOI of 0.1, we observed that all three MRSA strains lysed within 2–3 hours following phage infection but recovered high cell density after passaging one percent of the culture into fresh media. We therefore hypothesized that predation by Φ Staph1N selected for mutations conferred resistance against Φ Staph1N infection, allowing the culture to regrow. Indeed, Φ Staph1N was unable to form plaques on recovered MRSA cultures that survived in the initial Φ Staph1N infection ([Figure 1B](#)).

Next, we asked whether phage resistance against Φ Staph1N infection could cause a trade-off in drug resistance in MRSA. Since both phage and β -lactams interface with the bacterial cell wall, we hypothesized that phage resistance mutations could impact β -lactam activity. Parental strains of MRSA252, MW2, and LAC displayed higher minimal inhibitory concentrations (MICs) of ≥ 48 μ g/mL against the β -lactams oxacillin (OXA), cefazolin (CEF), amoxicillin (AMX), and amoxicillin & clavulanic acid (AMX+CA), visually indicated by their ability to form lawns surrounding antibiotic strips ([Figure 1C](#), [Table S2](#)). The MRSA strains were sensitive to vancomycin (VANC; MICs = 1.5 μ g/mL), which inhibits cell wall synthesis through a different mechanism than β -lactams.

By contrast, all strains that survived Φ Staph1N infection displayed a reduction in resistance against OXA, CEF, and AMX+CA, with fold reductions in MIC between 10 and 1000-fold ([Figure 1D](#)). No changes in MIC were observed with VANC. These observations suggest that the loss of resistance might be specific to the mechanism of action of β -lactams. Furthermore, phage-treated strains remained resistant to AMX alone, likely caused by the activity of β -lactamases present in the three MRSA strains. Together, these results suggest that predation by Φ Staph1N causes three different MRSA strains to lose their resistance towards β -lactams.

Next, we asked how infection at lower MOIs with Φ Staph1N affected β -lactam resistance in MRSA. We infected the three MRSA strains with Φ Staph1N at MOIs ranging from 0.01 to 10^{-5} , isolated the surviving MRSA cells, and tested the survivors for their MIC against oxacillin; each condition was tested in three independent replicates ([Table S2](#)). For MRSA252, we observed a ~3-order of magnitude fold reduction of MIC in the surviving cells at an MOI of 10^{-5} . With MW2, the reduction of MIC markedly decreased with lower phage levels, showing on average no reduction in MIC at

MOIs of 10^{-3} or lower (**Table S2**). For LAC, two of the replicates displayed an order of magnitude reduction of MIC at an MOI of 10^{-4} , while one replicate did show a strong decrease in MIC at the same MOI. These results suggest that for MRSA252, MOIs as low as 10^{-5} of Φ Staph1N infection can still drive the loss of resistance, while for MW2 and LAC, higher MOIs of phage are needed to ensure the same outcome of reduced β -lactam resistance.

We then asked whether the reduction in β -lactam can be achieved with a phage from a different taxonomical group. The MRSA LAC strain is sensitive to infection by ϕ INM1 ψ 6, a lytic version of the temperate phage ϕ INM1, displaying plaquing comparable to Φ Staph1N (**Table S1**, **Figure S1**). ϕ INM1 ψ 6 belongs to the siphovirus family and was derived from the *S. aureus* Newman strain.^{22–24} We therefore examined whether infection with ϕ INM1 ψ 6 could also reduce β -lactam resistance in LAC. We infected LAC with ϕ INM1 ψ 6 at an MOI of 0.1 and measured the MIC against oxacillin of the surviving cells. Here, recovered LAC cultures did not show a reduction in MICs (**Figure S3**). This suggests that ϕ INM1 ψ 6 is unable to reduce resistance in susceptible MRSA hosts, raising the possibility that different phages vary in their ability to change β -lactam resistance in MRSA.

Discovery of mutant in Φ Staph1N with enhanced activity against MRSA

A key advantage of phage therapy over conventional pharmaceutical antibiotics is that phages themselves can evolve. For Φ Staph1N, we noticed that while the phage was able to plaque all three MRSA strains, its plaque-forming efficiency was reduced on the MW2 and LAC strains (**Figure 1B**, **Figure S1**). Φ Staph1N plaques on MW2 and LAC strains were hazy and the overall efficiency of plaquing was approximately two orders of magnitude less than that with MRSA252. Unexpectedly, we consistently observed smaller, clear plaques arising in the larger, hazy plaques of LAC (**Figure 2A**). This led us to hypothesize that these clear plaques are caused by a mutant Φ Staph1N that plaques more efficiently on LAC. We therefore isolated phage clones from these single plaques and tested for their activity against MRSA. This mutant phage, which we called Evo2, is able to plaque on LAC and MW2 strains with higher efficiency, displaying comparable plaquing to MRSA252 (**Figure 2A**, **Figure S1**). In growth curve experiments, we further observed that Evo2 lyses MRSA cultures at lower MOIs compared to Φ Staph1N (**Figure S2**). Notably, Evo2 exhibits lytic activity against MW2 and LAC even at an MOI of 0.0001, a concentration at which Φ Staph1N does not show any detectable activity against the two strains.

We sequenced the genome of Evo2 to determine the genetic mechanism driving this enhanced activity. We observed a single point mutation in ORF141 that induces a premature stop codon (**Figure S4**). Sequence analysis with HHpred predicts ORF141 to be a putative DNA binding protein with an HTH motif (PDB: 2LVS, E-value: 2.5×10^{-9}). We speculate that this protein is a transcriptional regulator that when inactivated by a nonsense mutation, increases Φ Staph1N infectivity. Future studies will center on determining the mechanism of this mutation and why Evo2 only evolved in the LAC strain. Given Evo2's enhanced activity against MRSA, we then asked how predation by Evo2 impacted β -lactam resistance in our three MRSA strains. We infected MRSA252, MW2, and LAC with Evo2 at an MOI of 0.1 and measured the MICs against β -lactams after 48 hours of passaging. Broadly, similar to the Φ Staph1N infection, infection by Evo2 reduced the MICs of the three MRSA strains against OXA, CEF, and AMX+CA, while MICs against AMX alone and VAN did not change significantly (**Figure 2B**).

We then tested how different MOIs of Evo2 impacted β -lactam resistance in MRSA (**Table S2**). As with Φ Staph1N, we infected the three MRSA strains with Evo2 at varying MOIs from 10^{-2} to 10^{-5} and measured the MIC against oxacillin in the evolved MRSA; each condition was tested in three independent replicates. For most of the MRSA252 cultures, we did not see a significant difference in the fold reduction of MIC between Φ Staph1N and Evo2. Across all three strains, we observed that replicate cultures across different MOIs were unable to recover growth following Evo2

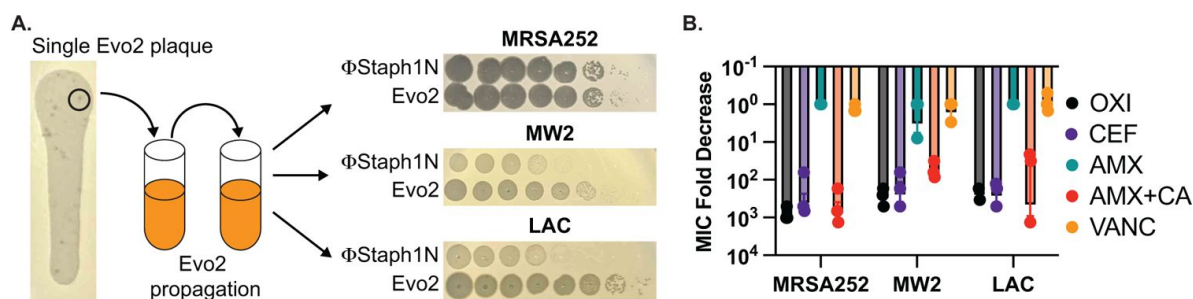


Figure 2.

Evo2 is a variant of ΦStaph1N with higher activity against MRSA.

A. Single Evo2 plaques appear inside hazy ΦStaph1N plaques on LAC (Left-most panel). Compared to ΦStaph1N, Evo2 shows comparable infectivity towards MRSA252 but improved infectivity towards MW2 and LAC (Right-most panels). **B.** Similar to ΦStaph1N, Evo2 infection reduces β-lactam resistance in MRSA.

infection (**Table S2**). However, cultures of MW2 and LAC that did regrow displayed a loss of oxacillin resistance. Thus, overall Evo2 displays a higher infectivity against the three MRSA strains and a greater potency in reducing β -lactam resistance.

Genomic mutations in MRSA strains following phage infection

Following these MIC tests, we hypothesized that the phage-treated MRSA strains evolved mutations that increased their sensitivity to β -lactams. We sequenced the genomes of three clonal isolates (A-C) from each MRSA strain that underwent Φ Staph1N, Evo2, or a mock infection. We observed that each MRSA strain evolved distinct mutation profiles (**Figure 3A**). Irrespective of the strain and phage treatment, most mutations were predicted to result in substitutions in the protein product, followed by truncations (**Figure S5**). Cluster of Orthologous Genes (COG) variants associated with transcription, cell wall/membrane/envelope biogenesis, coenzyme transport and metabolism, and defense mechanisms, were the most commonly found categories. Mutations in annotated genes that appeared at least twice across the clonal replicates are summarized in **Table 1**. Information on all detected genetic variants is listed in Supplementary Table 1.

In all MRSA strains, we found mutations that could account for the loss in β -lactam resistance. In MRSA252, both Φ Staph1N and Evo2 infection were selected for frameshift or nonsense mutations in the *femA* gene that would inactivate the protein product. FemA is required for the synthesis of the pentaglycine branch on *S. aureus* Lipid II, the peptidoglycan precursor (**Table 1**). Deletions of *femA* have been shown to increase susceptibility to β -lactams even when PBP2a (encoded by *mecA*) is expressed.^{37,38} These results thus provide a potential mechanism for how some MRSA252 cells lose β -lactam resistance after phage selection. At the same time, we noted the presence of other uncharacterized mutations in phage-treated MRSA252. For example, clone A of Φ Staph1N-treated cells carried 2 mutations: a frameshift in *femA* and a substitution mutation in an uncharacterized protein; meanwhile, clone B displayed a substitution mutation in *pfkA*, a predicted ATP-dependent 6-phosphofructokinase and mutation in an intergenic region; clone C showed a substitution mutation in a putative transport protein, called *yueF* (**Figure 3B**, Supplementary Table 1). The role of these mutations in mediating phage resistance or β -lactam sensitivity, if any, remains unknown.

In MW2, we found mutations in two transcriptional regulators, *mgrA* and *sarA* (**Figure 3A**, **Table 1**). Both *mgrA* and *sarA* belong to the family of MarR (multiple antibiotic resistance regulator)/SarA (staphylococcal accessory regulator A) proteins, which regulate drug resistance and virulence in *S. aureus*.^{25–27} In Φ Staph1N-treated MW2, only *mgrA* was mutated, while in Evo2-treated MW2, clones also showed nonsense mutations in *sarA*. We also found mutations in *metK* and *ytqA* which are both predicted to be associated with S-adenosylmethione: *metK* synthesizes SAM, while *ytqA* belongs to the radical SAM enzyme family and is predicted to be involved in tRNA modification.^{35,40} Phage-treated MW2 also displayed mutations in other genes, including *tcaB* and *murE*. Notably, each clonal replicate had multiple mutations in the genome, while by contrast untreated MW2 cells only displayed a deletion in an intergenic region that is not present in any of the phage-treated samples.

In the LAC strain, we observed a third, distinct mutational pattern (**Figure 3A**). Of note, we found nonsense mutations in *arlR*, which is part of the *arlRS* two-component signaling system (**Table 1**). The activity of ArlRS has been implicated in *S. aureus* virulence, pathogenicity, and oxacillin resistance.^{30,31} Further, we observed substitution mutations in *spoVG*, which is a transcription factor regulating the expression genes involved in a variety of functions, including cell wall metabolism (**Table 1**).³² Indeed, SpoVG activates the expression of *femA*.³³ Studies have shown that SpoVG modulates β -lactam antibiotic resistance by modulating cell wall synthesis.³³ Similar to the other two MRSA strains, phage-treated clones of LAC showed multiple mutations in their genomes, although their association with β -lactam sensitivity remains unclear. Finally, we observed mutations in *prsA* and *bioA* that appeared in both the mock and phage treatment conditions, suggesting that these mutations do not arise due to phage selection.

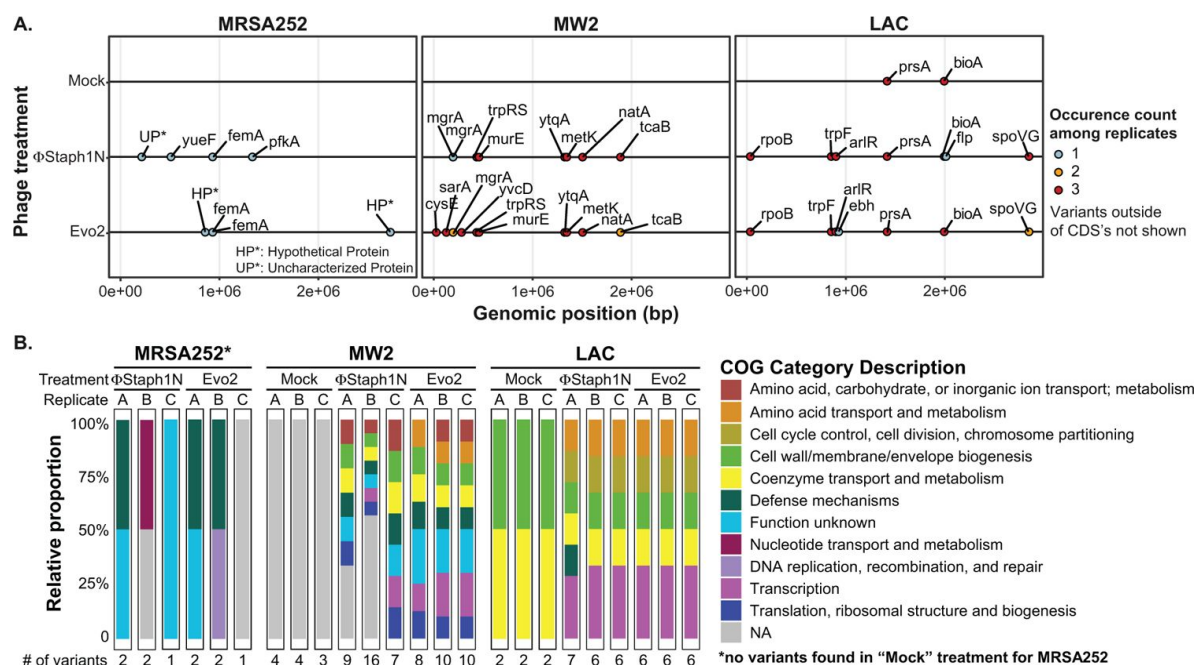


Figure 3.

Phage infection of MRSA strains produces distinct mutational profiles.

A. Coding sequences (CDS) with mutations from the three MRSA strains following phage treatment or mock treatment. For each strain, three isolates were sequenced and their mutations identified. Mutations are color-coded based on the number of occurrences among the three replicates. **B.** Categories of genes with mutations that arose in each MRSA strain and treatment condition.

Gene	Description	Strain	Phage infection	Reference
<i>sarA</i>	Transcriptional regulator of antibiotic resistance and virulence	MW2	Evo2	25,26
<i>mgrA</i>	Transcriptional regulator of antibiotic resistance and virulence	MW2	ΦStaph1N, Evo2	27,28
<i>rpoB</i>	Transcriptional regulator of antibiotic resistance	LAC	ΦStaph1N, Evo2	29
<i>arlR</i>	Transcriptional regulator of antibiotic resistance and virulence	LAC	ΦStaph1N, Evo2	28,30,31
<i>spoVG</i>	Transcriptional regulator of antibiotic resistance and virulence	LAC	ΦStaph1N, Evo2	32,33
<i>cysE</i>	Cysteine and methionine synthesis, serine O-acetyltransferase	MW2	Evo2	34
<i>metK</i>	Cysteine and methionine synthesis, S-adenosylmethionine (SAM) synthetase	MW2	ΦStaph1N, Evo2	35
<i>trpF</i>	Phenylalanine, tyrosine and tryptophan synthesis, phosphoribosylanthranilate isomerase	LAC	ΦStaph1N, Evo2	36
<i>femA</i>	Peptidoglycan synthesis, pentaglycine synthesis	MRSA252	ΦStaph1N, Evo2	37,38
<i>murE</i>	Peptidoglycan synthesis, UDP-MurNAc tripeptide synthesis	MW2	ΦStaph1N, Evo2	39
<i>trpS</i>	Aminoacyl-tRNA synthesis, tryptophanyl-tRNA synthesis	MW2	ΦStaph1N, Evo2	
<i>ytqA</i>	tRNA modifications, mnm ⁵ s ² U synthesis	MW2	ΦStaph1N, Evo2	40
<i>yvcD</i>	Unknown	MW2	Evo2	
<i>natA</i>	ABC transporter	MW2	ΦStaph1N, Evo2	41
<i>tcaB</i>	Predicted multidrug efflux pump	MW2	ΦStaph1N, Evo2	42

Table 1.

Mutated genes in MRSA following infection with phages ΦStaph1N or Evo2

Phage-treated MRSA strains display reduced virulence phenotypes

S. aureus strains encode a wide variety of virulence factors that facilitate bacterial infection, and previous studies have used MW2 and LAC strains as models of MRSA virulence.^{27,28} In MW2 and LAC, we observed the evolution of nonsense mutations in known virulence regulators, such as *mgrA*, *arlR*, and *sarA*. We therefore hypothesized that phage-treated MRSA cells could also display altered virulence phenotypes in addition to reduced β -lactam resistance. We first tested the ability of MRSA strains that survived phage predation for their ability to form biofilms in a Crystal Violet assay. We found that Evo2 infection of MRSA252 resulted in a significant reduction in Crystal Violet absorption compared to the parental strain. However, we show no significant difference in Crystal Violet absorption between parental, mock- and phage-treated MW2 and LAC strains (Figure S6).

Next, we tested whether phage infection could affect the hemolysis of rabbit blood cells. Hemolysis is mediated by the secretion of toxins, such as alpha toxins, and plays an important role in MRSA infection.⁴³ Expression of these toxins is regulated by virulence pathways that comprise numerous transcription factors, including *mgrA*, *arlR*, and *sarA*. Parental MW2 and LAC colonies lysed rabbit blood cells on blood agar plates, producing distinct halos of clearance around the bacterial cells. As expected, for untreated LAC, the total area of hemolysis was on average 3-fold larger than that of untreated MW2 (~210 mm² vs ~70 mm², respectively); with MRSA252, by contrast, lysis was not detected (Figure 4A). Following treatment with Φ Staph1N, we observed that MW2 and LAC displayed a reduced area of hemolysis by 4 to 5-fold. With Evo2-treated cells, we found that in MW2 the fold reduction was comparable to that of Φ Staph1N-treated cells. However, for Evo2-treated LAC, loss of hemolysis was even more pronounced, with two of the replicates showing no detectable hemolysis.

S. aureus is also able to bind to fibrinogen, resulting in the agglutination, or clumping, of bacteria. Clumping is thought to have several functions in the context of staphylococcal infections. Clumps are likely to be more resistant to clearance by the immune system, partly because they may be too large to be phagocytosed by neutrophils.²⁸ Similar to the expression of hemolysins, previous work has shown that clumping in MW2 and LAC is regulated by a network of virulence regulators. For MW2 and LAC, we observed that phage-treated cells displayed less clumping than the mock-treated or parental strain. For MW2, Φ Staph1N infection resulted in a modest reduction, while Evo2 infection resulted in a reduction of approximately 3-fold (Figure 4B). In LAC, we found that both Φ Staph1N and Evo2 treatment resulted in comparable reductions of clumping in surviving cells.

Discussion

Here we report the discovery that infection by the myovirus Φ Staph1N and its derivative Evo2 can drive MRSA populations to evolve favorable genetic trade-offs between phage and β -lactam resistance. MRSA strains that survived phage infection were resistant to phage; however, they displayed up to a 1000-fold decrease in their MICs against β -lactam antibiotics. Sequencing revealed that each of the three MRSA strains tested evolved distinct mutational profiles following phage infection. Of these mutations, a significant portion of the affected genes are involved in transcriptional regulation, antibiotic resistance, and cell maintenance. Concurring with the sequencing data, phage-treated MRSA strains displayed attenuated virulence phenotypes in the form of reduced hemolysis and cellular agglutination.

Our system highlights the multitude of evolutionary paths that different bacterial strains can undertake to evolve resistance against phage. Most staphylococcal phages are thought to adsorb to *S. aureus* by recognizing wall teichoic acid (WTA). Prior work has revealed that phage resistance in

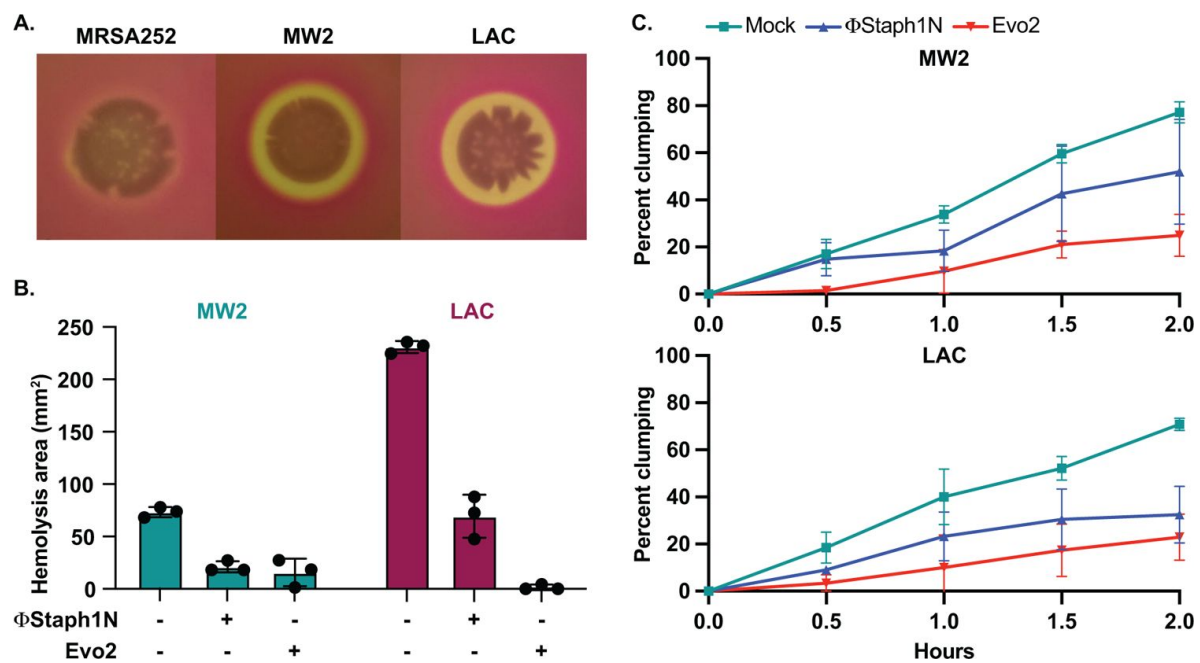


Figure 2.

Phage treatment of MRSA results in attenuated virulence phenotypes.

A. MW2 and LAC strains display hemolytic activity on rabbit blood agar plates, while MRSA252 does not. **B.** Phage-treated MW2 and LAC strains display reduced hemolysis compared to uninfected cells. **C.** Surviving cultures of MW2 and LAC treated with either ΦStaph1N (blue) or Evo2 (red) show reduced clumping rates compared to mock untreated cells (teal).

S. aureus can arise from the disruption of genes involved in the synthesis and modification of WTA, such as *tagO* or *tarM*.⁴⁴ Some MRSA strains also alter cell wall glycosylation through dedicated genes encoded on prophages.¹¹ Additionally, the presence of cell wall-bound Protein A has been shown to decrease phage absorption, likely by masking WTA.⁴⁵ However, our results suggest that the evolution of phage resistance can potentially impact networks of genes in the different MRSA strains, including metabolic genes and transcription factors. For example, both *mgrA* and *arlRS* regulate the expression of proteins associated with the cell wall, such as the Giant Staphylococcal Surface Protein (GSSP).²⁸ We hypothesize that mutations in these regulators might perturb the composition of proteins in the cell wall, which in turn can disrupt phage binding or prevent injection of the phage nucleic acid into the cytoplasm. Further studies are needed to dissect the effect of these gene networks on phage resistance and β -lactam resistance.

Our study also shows some of the challenges associated with determining phage/antibiotic resistance trade-offs. Not only do different bacterial strains evolve distinct mutations, but individual clones can accumulate multiple mutations in their genome. Such complex dynamics have been observed in other coevolving bacteria-phage systems before and can serve as useful models to study evolution. However, they make it difficult to predict broader phenotypic outcomes in the bacteria following phage infection. For example, phage resistance has also been shown to confer so-called “trade-ups,” where a non-selected trait is enhanced (e.g. cross-resistance between phage and antibiotic). In our study, we cannot rule out that our phage-treated MRSA has evolved (or will evolve) trade-ups. Although further research is required to test this possibility, the potential clinical benefit of reduced resistance and virulence in MRSA may outweigh the risks associated with trade-ups.

Despite evolving distinct mutational profiles, the three MRSA strains nonetheless displayed a convergence of phenotypes in the form of enhanced phage resistance, reduced β -lactam resistance, and attenuated virulence. We posit that this convergence is caused by the involvement of the cell wall in all three phenotypic outcomes. Phages must interface with the cell wall to adsorb, β -lactams target proteins associated with the maintenance of the cell wall, and many *S. aureus* virulence factors are embedded within the cell wall or need to get secreted through it. The integrity and composition of the cell wall can be determined by numerous genes, ranging from single proteins involved in cell wall synthesis, such as *femA*, to transcriptional regulators that control the expression of cell wall-associated proteins, such as *mgrA*. Thus, the mutational patterns observed in each MRSA strain might reflect unique genetic solutions that enable the bacteria to adapt to the phage predation, while also maximizing the fitness for that particular strain.

From a biomedical perspective, exploiting phage/antibiotic resistance trade-offs have been proposed as a means to combat resistance in bacterial pathogens.¹⁵ Not only could phage treatments reduce the bacterial load of an infection, but also potentially resensitize the bacterial population to antibiotics that it was previously resistant against. For example, in our study, we observed that the MIC of phage-treated MRSA against oxacillin dropped below 2 μ g/mL, which is designated by the FDA as the oxacillin MIC susceptible breakpoint for *S. aureus*. We speculate that infection by Φ Staph1N, Evo2, or an equivalent phage can facilitate the re-deployment of β -lactams in the treatment against MRSA infections. Additionally, phage-treated MRSA might also lose virulence and become more susceptible to immune clearance, which can further improve treatment outcomes. Future studies will need to examine this hypothesis in animal infection models.

Acknowledgements

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Online Methods

Strains and culture conditions

The bacterial strains used in this study are listed in [Table S1](#). Unless otherwise indicated, all MRSA strains (LAC, MRSA252, MW2) were grown in Brain Heart Infusion (BHI) media at 37 °C with shaking (235 RPM).

Plate-based plaque assay

Bacterial lawns were prepared by mixing 100 µL of an overnight culture with 5 mL of melted BHI agarose (top agar). The bacteria and top agar mixture were poured onto a solid BHI plate. The plate was dried for 10 minutes. 10-fold serial dilutions (10^0 - 10^{-7} unless otherwise noted) of phage were then spotted on the bacterial lawn. Plates were then incubated at 37 °C for 16 hours. Phage titer in plaque-forming units per µL (pfu/µL) was then calculated.

Phage infection assay

MRSA strains were plated onto BHI agar plates and grown overnight. Individual colonies from the parental strains (also referred to as P0) were picked. Single colonies were inoculated in a round bottle tube containing 5 mL BHI broth. The cultures were incubated at 37 °C, 235 RPM for 24 hours. The grown P0 cultures were then diluted 1:100 into fresh 5 mL BHI broth. At the early log phase (OD ~0.3), the bacterial cultures were treated with phage at an MOI of 0.1, unless indicated otherwise. The treated bacterial cultures were incubated at 37 °C with shaking (235 RPM) for 24 hours. The cultures were then passaged 1:100 into fresh 5 mL BHI broth. This passage was then grown at 37 °C, 235 RPM, for another 24 hours. Surviving cultures were then used for both phenotypic assays and sequencing experiments. As a negative control, MRSA strains were passaged using the steps described above without phage treatment (mock).

MIC assay

Bacterial lawns were normalized to contain 1×10^8 CFU/mL bacteria mixed with top agar for a total volume of 5 mL. The bacteria and top agar mixture is poured onto a solid BHI plate. The plate was dried for 10 minutes. MIC with increasing concentrations of antibiotics were placed on the semi-dried bacterial lawn and allowed to dry for 10 minutes. The plates are then incubated at 37 °C overnight. For analysis, the plates were imaged, and the MIC of the bacteria was determined. The MIC is determined at the edge of the inhibition ellipse intersects the side of the strip.

Rabbit blood hemolysis

Phage-treated or mock-treated cultures were diluted to an OD600 of 0.1, 5 µL of this dilution was spotted on rabbit blood TSB agar plates and incubated at 37 °C for 24 hours. The area of clearance was determined by the following formula: $[\pi (\text{diameter of clearance}/2)^2] - [\pi (\text{diameter of bacterial spot}/2)^2]$

Clumping assay

Clumping assays were performed as described previously.⁴⁶ In short, overnight cultures were diluted 1:100 and incubated at 37 °C until the cultures reached an OD₆₀₀ of 1.5. At this point, 1.5 mL of culture was washed two times and resuspended with PBS. Lyophilized human plasma was added for a final concentration of 1.25%. Resuspended cells were left to sit statically at room temperature. 100uL were taken from the top of the cell suspensions in 30-minute intervals and the OD₆₀₀ was measured.

Biofilm assay

Biofilm assay was performed using the crystal violet method as outlined.⁴⁷ In brief, overnight cultures grown in BHI at 37 °C were back diluted 1:10 into a 96-well bottomed microwell plate. The plates were incubated at 37 °C overnight. The contents in the plate were discarded and washed with PBS. Biofilm fixation was done with the addition of sodium acetate (2%). Crystal violet (0.1%) was used for staining followed by a final wash with PBS. Absorbance at 600 nm was read using a spectrophotometer.

DNA sequencing and genome assembly

Following published protocols, genomic DNA from bacteria and phage was isolated using phenol-chloroform extraction. Purified DNA was sent to Plasmidsaurus and SeqCenter for Nanopore and Illumina sequencing, respectively. Reference genomes for bacterial strains were assembled using Flye v2.9.3 with default settings for long-reads. This resulted in 1 singular contig assemblies for 252 (2902592bp, 125x) and MW2 (2820460, 600x), and 3 contigs for LAC (2907712, 645x). Phage assemblies for Evo2 and ΦStaph1N were done with the SPAdes assembler v3.15.5.

Open-reading frames (ORFs) were called on the assembled bacterial genomes using Prodigal v.2.6.3,⁴⁸ resulting in a gene-feature file (GFF), and translated genes as .faa and .fna formats. We used BLASTp (Accessed June 4th, 2024), against the protein BLAST database swissprot_2023-06-28, with an expectation value cutoff of 0.001. The top hit for each ORF was used as the final functional annotation.

Mutation identification

A total of 27 genomic samples were collected. DNA was extracted and sent for long-read sequencing using Oxford Nanopore Technology (Plasmidsaurus, San Francisco, USA). Reads were filtered using filtlong v0.2.1 using default settings (<https://github.com/rrwick/Filtlong>) and with the -p flag 95 (keeping 95% of the best reads). Quality-filtered long reads were mapped against the respective genomes using minimap2 2.22-r1101,⁴⁹ resulting in 1 alignment file output per sample (.sam file). The read mapping software Minimap2⁴⁹ was selected because of its suitability to map long reads. Samtools v1.20⁵⁰ was used to convert the .sam files into .bam files, sort the bam file, index the bam files, and generate a coverage table for each position along the alignment.

Reference files (fasta and GFF files) and the 53 “sorted.bam” alignments files were imported into Geneious. Variant calling was performed using Geneious Prime Geneious Prime ® 2024.0.2 (www.geneious.com), using the custom settings: 10% coverage, minimum 95% variant frequency threshold, and the option for “Analyze effect of variants on translation” checked. Variant results and genome annotations table files were exported as a tab-separated-table, and visualized using R v4.4.0, mostly with the tidyverse package. Sequencing data processing, quality filtering, and mapping were performed at the Center for High-Throughput Computing (<https://chtc.cs.wisc.edu/>). BLASTp was performed on usegalaxy.eu (Accessed June 4th, 2024).

Supplemental data

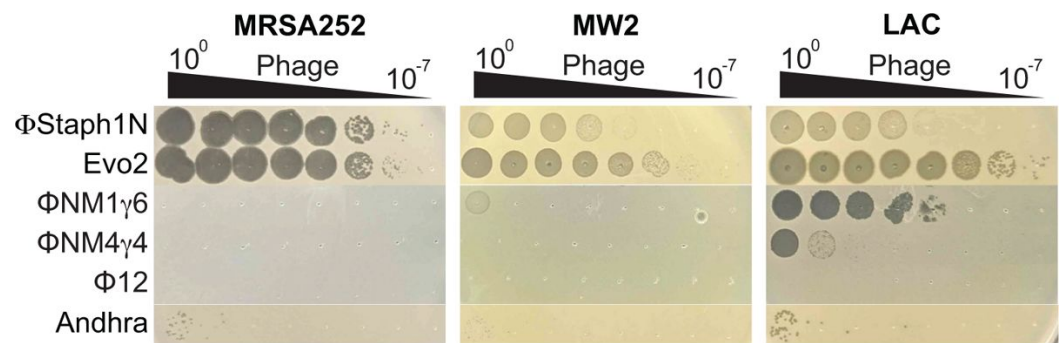


Figure S1.

Phage sensitivity of MRSA strains.

Efficiencies of plaquing of phages on MRSA252, MW2, and LAC. Phages were 10-fold serially diluted and spotted onto top agar overlays of each strain. Phage Evo2 shows the highest plaquing efficiency on all three strains.

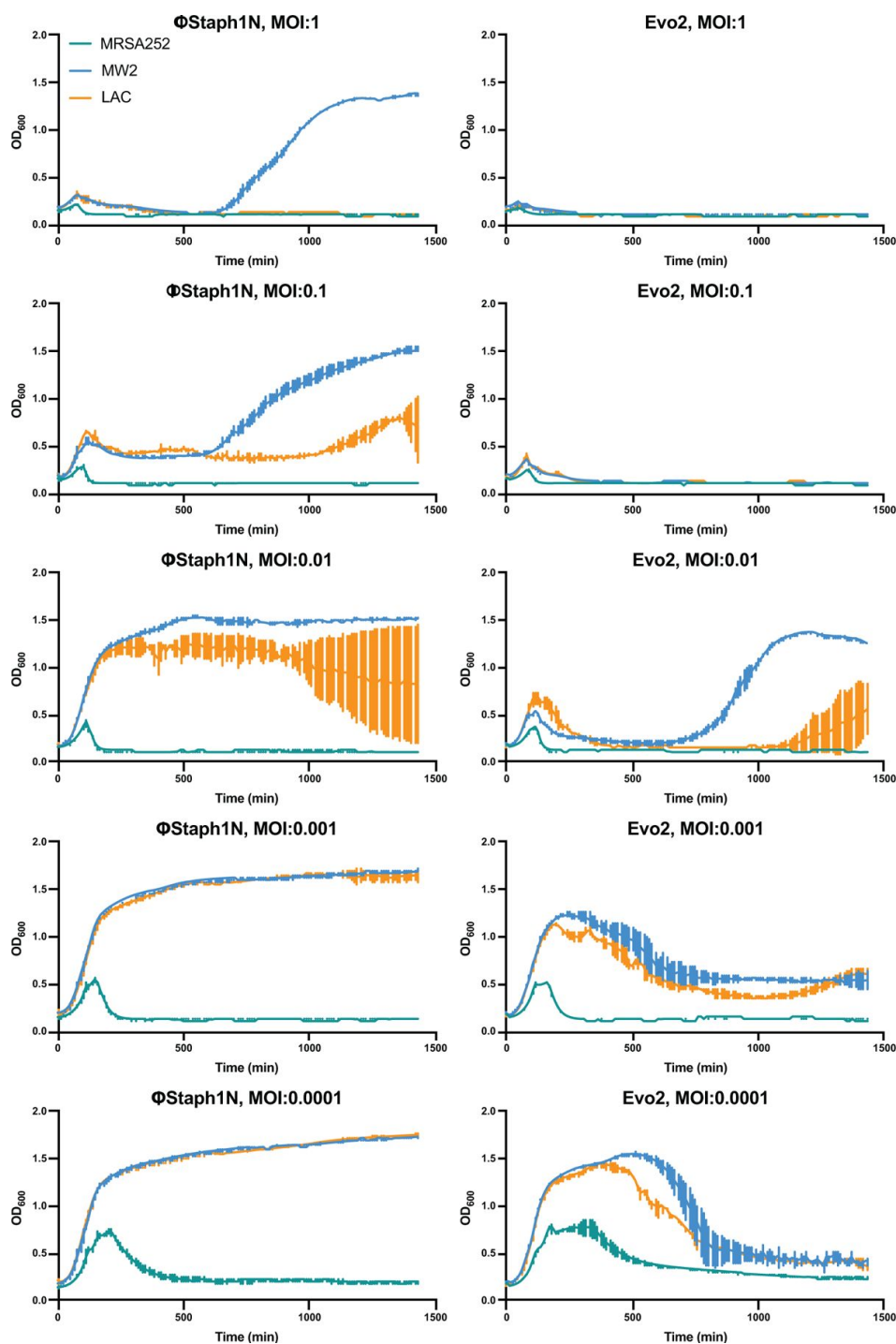


Figure S2.

Growth curves of MRSA strains under varying levels of phage infection.

MRSA252, MW2, and LAC cultures were infected with either Φ Staph1N or Evo2 at the indicated multiplicity of infection (MOI). The optical density (OD_{600}) of the cultures was monitored on an automated plate reader. Each condition was tested in 3 independent replicates.

Phage λ NM1 ψ 6 infection LAC does not drive the loss of β -lactam resistance.

Oxacillin		Cefazolin		Amoxicillin		Amoxicillin + clavulanic acid		Vancomycin	
Mock	ΦStaph1N	Mock	ΦStaph1N	Mock	ΦStaph1N	Mock	ΦStaph1N	Mock	ΦStaph1N
									

Sequencing analysis of Evo2.

ΦStaph1N T A C T A T A A A G T A G G T A A G C G T A A T A A G A A A T C A C C T G T A G A A
 Evo2 T A C T A T A A A G T A G G T A A G C G T A A T A A G A A A T A A C C T G T A G A A
 Tyr Tyr Lys Val Gly Lys Arg Asn Lys Lys Ser/Stop Pro Val Glu
 70 80
 Residue position in ORF141

Figure S5.

Types of polymorphisms in MRSA strains following infection by phage or a mock treatment.

Plotted are the polymorphisms that were found in a gene with an assigned COG category.

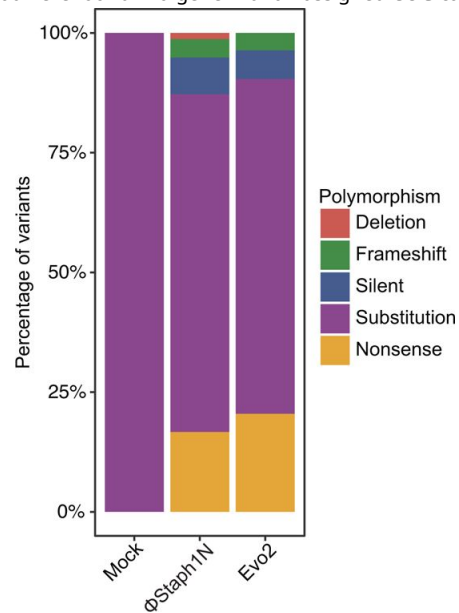
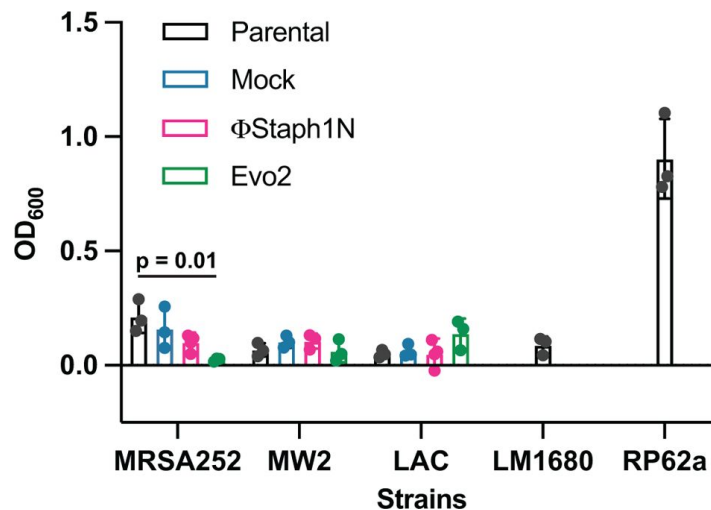


Figure S6.

Effect of phage infection on biofilm formation in MRSA strains.

Cultures were infected or mock-infected with either ϕStaph1N or Evo2. RP62a is a strain of *S. epidermidis* with known biofilm forming capability, while LM1680 is a derivative of RP62a that has lost biofilm forming ability.^{51,52} Biofilm biomass was assessed by staining with Crystal Violet. Solubilized crystal violet was quantified by measuring absorbance at 600 nm. Values represent averages and standard deviations of three replicates. Statistical significance was determined with a two-tailed t-test.



Strain Name	Comments	Source or Reference
MRSA MW2 (USA400)		Baba <i>et al.</i> , 2002
MRSA LAC (USA300)		Voyich <i>et al.</i> , 2005
MRSA252 (USA200)		Holden <i>et al.</i> , 2004
<i>S. epidermidis</i> RP62a	methicillin-resistant biofilm-producing <i>S. epidermidis</i>	Christensen <i>et al.</i> , 1982
<i>S. epidermidis</i> LAM1680	Derived from <i>S. epidermidis</i> RP62a; carries genomic deletion that inactivates biofilm production	Jiang <i>et al.</i> , 2016
ΦStaph1N		Łobocka <i>et al.</i> , 2012
Evo2	Derived from ΦStaph1N	This study
ΦNM1γ6	Lytic version of ΦNM1	Marraffini laboratory
ΦNM4γ4	Lytic version of ΦNM4	Marraffini laboratory
Φ12		Marraffini laboratory
Andhra	Infects <i>S. epidermidis</i>	Hatoum-Aslan laboratory

Table S1.

Bacterial strains and bacteriophages used in this study

	MRSA252					
	Staph1N			Evo2		
MOI	Rep 1	Rep 2	Rep 3	Rep 1	Rep 2	Rep 3
10 ⁻²	0.25	0.125	0.38	NG	2	0.5
10 ⁻³	NG	0.94	0.19	1	0.75	1
10 ⁻⁴	0.5	0.25	0.19	0.75	1	0.5
10 ⁻⁵	0.25	0.38	NG	0.38	NG	NG
Mock	>256	>256	>256	>256	>256	>256
	MW2					
	Staph1N			Evo2		
MOI	Rep 1	Rep 2	Rep 3	Rep 1	Rep 2	Rep 3
10 ⁻²	3	24	24	4	NG	NG
10 ⁻³	32	24	48	4	NG	NG
10 ⁻⁴	48	96	32	3	NG	NG
10 ⁻⁵	96	64	24	2	NG	NG
Mock	96	48	32	96	48	32
	LAC					
	Staph1N			Evo2		
MOI	Rep 1	Rep 2	Rep 3	Rep 1	Rep 2	Rep 3
10 ⁻²	NG	NG	2	0.064	NG	NG
10 ⁻³	NG	3	1.5	0.032	NG	NG
10 ⁻⁴	32	1.5	1	NG	NG	NG
10 ⁻⁵	32	16	0.38	NG	NG	NG
Mock	32	48	48	32	48	48

NG: no growth detected

Table S2.

MIC against oxacillin of MRSA strains treated with different MOIs of phage NG: no growth detected

References

1. Gorwitz R. J., et al. (2008) **Changes in the Prevalence of Nasal Colonization with *Staphylococcus aureus* in the United States, 2001–2004** *J INFECT DIS* **197**:1226–1234
2. Tomasz A (1979) **The Mechanism of the Irreversible Antimicrobial Effects of Penicillins: How the Beta-Lactam Antibiotics Kill and Lyse Bacteria** *Annu. Rev. Microbiol* **33**:113–137
3. World Health Organization (2019) **Critically Important Antimicrobials for Human Medicine** *World Health Organization, Geneva*
4. Zapun A., Contreras-Martel C., Vernet T. (2008) **Penicillin-binding proteins and β -lactam resistance** *FEMS Microbiol Rev* **32**:361–385
5. Murray C. J. L., et al. (2022) **Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis** *The Lancet* **399**:629–655
6. Suttle S. W (1999) **Viruses and Nutrient Cycles in the Sea: Viruses play critical roles in the structure and function of aquatic food webs** *BioScience*
7. Kortright K. E., Chan B. K., Koff J. L., Turner P. E (2019) **Phage Therapy: A Renewed Approach to Combat Antibiotic-Resistant Bacteria** *Cell Host & Microbe* **25**:219–232
8. Hatoum-Aslan A (2021) **The phages of staphylococci: critical catalysts in health and disease** *Trends in Microbiology* **29**:1117–1129
9. Oechslin F (2018) **Resistance Development to Bacteriophages Occurring during Bacteriophage Therapy** *Viruses* **10**
10. Azam A. H., Tanji Y (2019) **Peculiarities of *Staphylococcus aureus* phages and their possible application in phage therapy** *Appl Microbiol Biotechnol* **103**:4279–4289
11. Gerlach D., et al. (2018) **Methicillin-resistant *Staphylococcus aureus* alters cell wall glycosylation to evade immunity** *Nature* **563**:705–709
12. Meyer J. R., et al. (2012) **Repeatability and Contingency in the Evolution of a Key Innovation in Phage Lambda** *Science* **335**:428–432
13. Bernheim A., Sorek R (2020) **The pan-immune system of bacteria: antiviral defence as a community resource** *Nat Rev Microbiol* **18**:113–119
14. Maynard N. D., et al. (2010) **A Forward-Genetic Screen and Dynamic Analysis of Lambda Phage Host-Dependencies Reveals an Extensive Interaction Network and a New Anti-Viral Strategy** *PLoS Genet* **6**
15. Oromí-Bosch A., Antani J. D., Turner P. E (2023) **Developing Phage Therapy That Overcomes the Evolution of Bacterial Resistance** *Annu. Rev. Virol* **10**:503–524

16. Berryhill B. A., Huseby D. L., McCall I. C., Hughes D., Levin B. R (2021) **Evaluating the potential efficacy and limitations of a phage for joint antibiotic and phage therapy of *Staphylococcus aureus* infections** *Proc. Natl. Acad. Sci. U.S.A* **118**
17. Voyich J. M., et al. (2005) **Insights into Mechanisms Used by *Staphylococcus aureus* to Avoid Destruction by Human Neutrophils** *The Journal of Immunology* **175**:3907–3919
18. Holden M. T. G., et al. (2004) **Complete genomes of two clinical *Staphylococcus aureus* strains: Evidence for the rapid evolution of virulence and drug resistance** *Proc. Natl. Acad. Sci. U.S.A* **101**:9786–9791
19. Kobayashi S. D., et al. (2011) **Comparative analysis of USA300 virulence determinants in a rabbit model of skin and soft tissue infection** *J Infect Dis* **204**:937–941
20. Xia G., Wolz C (2014) **Phages of *Staphylococcus aureus* and their impact on host evolution** *Infection, Genetics and Evolution* **21**:593–601
21. Deghorain M., Van Melder L (2012) **The Staphylococci Phages Family: An Overview** *Viruses* **4**:3316–3335
22. Mo C. Y., et al. (2021) **Type III-A CRISPR immunity promotes mutagenesis of staphylococci** *Nature* **592**:611–615
23. Goldberg G. W., Jiang W., Bikard D., Marraffini L. A (2014) **Conditional tolerance of temperate phages via transcription-dependent CRISPR-Cas targeting** *Nature* **514**:633–637
24. Baba T., Bae T., Schneewind O., Takeuchi F., Hiramatsu K (2008) **Genome Sequence of *Staphylococcus aureus* Strain Newman and Comparative Analysis of Staphylococcal Genomes: Polymorphism and Evolution of Two Major Pathogenicity Islands** *J Bacteriol* **190**:300–310
25. Li L., et al. (2016) **The Global Regulon *sarA* Regulates β -Lactam Antibiotic Resistance in Methicillin-Resistant *Staphylococcus aureus* In Vitro and in Endovascular Infections** *J Infect Dis* **214**:1421–1429
26. Zielinska A. K., et al. (2012) **SARA - mediated repression of protease production plays a key role in the pathogenesis of *S. taphylococcus aureus* USA 300 isolates** *Molecular Microbiology* **86**:1183–1196
27. Crosby H. A., et al. (2016) **The *Staphylococcus aureus* Global Regulator MgrA Modulates Clumping and Virulence by Controlling Surface Protein Expression** *PLoS Pathog* **12**
28. Kwiecinski J. M., et al. (2021) ***Staphylococcus aureus* uses the ArIRS and MgrA cascade to regulate immune evasion during skin infection** *Cell Reports* **36**
29. Panchal V. V., et al. (2020) **Evolving MRSA: High-level β -lactam resistance in *Staphylococcus aureus* is associated with RNA Polymerase alterations and fine tuning of gene expression** *PLoS Pathog* **16**
30. Bai J., et al. (2019) **The role of ArIRS in regulating oxacillin susceptibility in methicillin-resistant *Staphylococcus aureus* indicates it is a potential target for antimicrobial resistance breakers** *Emerging Microbes & Infections* **8**:503–515

31. Walker J. N., et al. (2013) **The *Staphylococcus aureus* ArlRS Two-Component System Is a Novel Regulator of Agglutination and Pathogenesis** *PLoS Pathog* **9**
32. Schulthess B., et al. (2011) **The σ B - Dependent yabJ-spoVG Operon Is Involved in the Regulation of Extracellular Nuclease, Lipase, and Protease Expression in *Staphylococcus aureus*** *J Bacteriol* **193**:4954–4962
33. Liu X., Zhang S., Sun B (2016) **SpoVG Regulates Cell Wall Metabolism and Oxacillin Resistance in Methicillin-Resistant *Staphylococcus aureus* Strain N315** *Antimicrob Agents Chemother* **60**:3455–3461
34. Chen C., et al. (2019) **Characterization of serine acetyltransferase (CysE) from methicillin-resistant *Staphylococcus aureus* and inhibitory effect of two natural products on CysE** *Microbial Pathogenesis* **131**:218–226
35. Markham G. D., DeParasis J., Gatmaitan J (1984) **The sequence of metK, the structural gene for S-adenosylmethionine synthetase in *Escherichia coli*** *Journal of Biological Chemistry* **259**:14505–14507
36. Proctor A. R., Kloos W. E (1973) **Tryptophan Biosynthetic Enzymes of *Staphylococcus aureus*** *J Bacteriol* **114**:169–177
37. Maidhof H., Reinicke B., Blümel P., Berger-Bächi B., Labischinski H (1991) **femA, which encodes a factor essential for expression of methicillin resistance, affects glycine content of peptidoglycan in methicillin-resistant and methicillin-susceptible *Staphylococcus aureus* strains** *J Bacteriol* **173**:3507–3513
38. Srisuknimit V., Qiao Y., Schaefer K., Kahne D., Walker S (2017) **Peptidoglycan Cross-Linking Preferences of *Staphylococcus aureus* Penicillin-Binding Proteins Have Implications for Treating MRSA Infections** *J. Am. Chem. Soc* **139**:9791–9794
39. Gardete S., et al. (2004) **Role of murE in the Expression of β -Lactam Antibiotic Resistance in *Staphylococcus aureus*** *J Bacteriol* **186**:1705–1713
40. Jaroch M., et al. (2024) **Alternate routes to mnm 5 s 2 U synthesis in Gram-positive bacteria** *J Bacteriol* **206**:e00452–23
41. Kobayashi K., et al. (2001) **Comprehensive DNA Microarray Analysis of *Bacillus subtilis* Two-Component Regulatory Systems** *J Bacteriol* **183**:7365–7370
42. Maki H., McCallum N., Bischoff M., Wada A., Berger-Bächi B (2004) **tcaA Inactivation Increases Glycopeptide Resistance in *Staphylococcus aureus*** *Antimicrob Agents Chemother* **48**:1953–1959
43. Otto M. (2014) ***Staphylococcus aureus* toxins** *Current Opinion in Microbiology* **17**:32–37
44. Jurado A., Fernández L., Rodríguez A., García P (2022) **Understanding the Mechanisms That Drive Phage Resistance in *Staphylococci* to Prevent Phage Therapy Failure** *Viruses* **14**
45. Moller A. G., et al. (2021) **Genes Influencing Phage Host Range in *Staphylococcus aureus* on a Species-Wide Scale** *mSphere*
46. Crosby H. A., Kwiecinski J. M., Horswill A. R. (2021) **In Vitro Assay for Quantifying Clumping of *Staphylococcus aureus*** in *Staphylococcus aureus* (ed. Rice, K. C.) **2341**:31–36

47. Feoktistova M., Geserick P., Leverkus M (2016) **Crystal Violet Assay for Determining Viability of Cultured Cells** *Cold Spring Harb Protoc* 2016
48. Hyatt D., et al. (2010) **Prodigal: prokaryotic gene recognition and translation initiation site identification** *BMC Bioinformatics* 11
49. Li H (2018) **Minimap2: pairwise alignment for nucleotide sequences** *Bioinformatics* 34:3094–3100
50. Li H., et al. (2009) **The Sequence Alignment/Map format and SAMtools** *Bioinformatics* 25:2078–2079
51. Christensen G. D., Simpson W. A., Bisno A. L., Beachey E. H (1982) **Adherence of slime-producing strains of Staphylococcus epidermidis to smooth surfaces** *Infect Immun* 37:318–326
52. Jiang W., et al. (2013) **Dealing with the Evolutionary Downside of CRISPR Immunity: Bacteria and Beneficial Plasmids** *PLoS Genet* 9

Editors

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Reviewer #1 (Public review):

Summary:

These authors have asked how lytic phage predation impacts antibiotic resistance and virulence phenotypes in methicillin-resistant *Staphylococcus aureus* (MRSA). They report that staphylococcal phages cause MRSA strains to become sensitized to β -lactams and to display reduced virulence. Moreover, they identify mutations in a set of genes required for phage infection that may impact antibiotic resistance and virulence phenotypes.

Strengths:

Phage-mediated re-sensitization to antibiotics has been reported previously but the underlying mutational analyses have not been described. These studies suggest that phages and antibiotics may target similar pathways in bacteria.

Weaknesses:

One limitation is the lack of mechanistic investigations linking particular mutations to the phenotypes reported here. This limits the impact of the work.

Another limitation of this work is the use of lab strains and a single pair of phages. However, while incorporation of clinical isolates would increase the translational relevance of this work it is unlikely to change the conclusions.

Reviewer #2 (Public review):**Summary:**

The work presented in the manuscript by Tran et al deals with bacterial evolution in the presence of bacteriophage. Here, the authors have taken three methicillin-resistant *S. aureus* strains that are also resistant to beta-lactams. Eventually, upon being exposed to phage, these strains develop beta-lactam sensitivity. Besides this, the strains also show other changes in their phenotype such as reduced binding to fibrinogen and hemolysis.

Strengths:

The experiments carried out are convincing to suggest such in vitro development of sensitivity to the antibiotics. Authors were also able to "evolve" phage in a similar fashion thus showing enhanced virulence against the bacterium. In the end, authors carry out DNA sequencing of both evolved bacteria and phage and show mutations occurring in various genes. Overall, the experiments that have been carried out are convincing.

Weaknesses:

Although more experiments are not needed, additional experiments could add more information. For example, the phage gene showing the HTH motif could be reintroduced in the bacterial genome and such a strain can then be assayed with wildtype phage infection to see enhanced virulence as suggested. At least one such experiment proves the discoveries regarding the identification of mutations and their outcome. Secondly, I also feel that authors looked for beta-lactam sensitivity and they found it. I am sure that if they look for rifampicin resistance in these strains, they will find that too. In this case, I cannot say that the evolution was directed to beta-lactam sensitivity; this is perhaps just one trait that was observed. This is the only weakness I find in the work. Nevertheless, I find the experiments convincing enough; more experiments only add value to the work.